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Analytical

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① * Gradient magnitude:

$$G = \sqrt{G_x^2 + G_y^2} = \sqrt{24^2 + 18^2} = \sqrt{576 + 324} = 30$$

* Gradient direction:

$$\theta = \tan^{-1} \left(\frac{dy}{dx} \right) = \tan^{-1} \left(\frac{18}{24} \right) = \tan^{-1} (0.75)$$

② Non-maximum Suppression.

given,

pixel	P_1	P_2	P_3	P_4	P_5
Magnitude	25	50	90	60	30

for a horizontal gradient direction, compare each pixel with its horizontal neighbors.

* $P_1 = 25$ Suppressed.* $P_2 = 50$ Suppressed because $50 < 90$ * $P_3 = 90$ Not suppressed because it is greater than 50 and 60* $P_4 = 60$ Suppressed because $60 < 90$ * $P_5 = 30$ Suppressed.Result:-

Pixel	P_1	P_2	P_3	P_4	P_5
After NMS	0	0	90	0	0

P_3 is the remaining edge pixel

③ Double Thresholding.

given:

* high threshold = 100

* low threshold = 50

Rules:

* Magnitude $\geq 100 \rightarrow$ strong edge.

* $50 \leq \text{Magnitude} < 100 \rightarrow$ weak edge

* Magnitude $< 50 \rightarrow$ non-edge.

Pixel	Magnitude	Classification
P_1	135	Strong edge
P_2	90	Weak edge
P_3	45	non edge
P_4	120	Strong edge
P_5	70	Weak edge
P_6	20	non edge.

④ Hysteresis Thresholding.

Given:

Pixel	magnitude	Classification.
P_1	130	Strong
P_2	95	Weak
P_3	40	No edge
P_4	115	Strong
P_5	65	Weak
P_6	25	No-edge
P_7	105	Strong.

Thresholding :-

High = 100

Low = 50

The

question

states

* P_2 is connected to any strong edge.

* P_5 is not connected to any strong edge

1) Strong edge

2) weak edge

3) non-edge

P_1, P_4, P_7

P_2, P_5

P_3, P_6

Single-edge pixels after Hysteresis.

* $P_1 \rightarrow$ strong $\rightarrow$ kept

* $P_2 \rightarrow$ weak but connected to $P_1 \rightarrow$ kept

* $P_3 \rightarrow$ non-edge $\rightarrow$ removed.

* $P_4 \rightarrow$ strong $\rightarrow$ kept.

* $P_5 \rightarrow$ weak and not connected to strong edge $\rightarrow$ remove

* $P_6 \rightarrow$ non-edge $\rightarrow$ removed

* $P_7 \rightarrow$ strong $\rightarrow$ kept.

Final edge pixels. = P_1, P_2, P_4, P_7

⑤ Hough Circle.

Given

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$P_1 = (5, 4) ; C_1 = (2, 4)$$

$$P_2 = (7, 6) ; C_2 = (4, 3)$$

For $C = (2, 4) P_1 (5, 4)$

$$r = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{9} = 3$$

$P_2 (7, 6)$

$$r = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{25+4} = \sqrt{29} \approx 5.39$$

$$\boxed{\approx 5.39}$$

$$C_2 = (4, 3) \quad P_1 = (5, 4)$$

$$r = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{2} \approx 1.41$$

$$P_2 = (7, 6)$$

$$r = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{9+9} = \sqrt{18} \approx 4.24$$

$$\boxed{1.41 \approx 4.24}$$

⑥ calculate the radius.

$$P(8, 6) \quad C(4, 3)$$

$$r = \sqrt{(8-4)^2 + (6-3)^2} = \sqrt{16+9} = 5$$

Radius = 5 units.

③ $P = (8, 8) \quad C = (5, 4)$

$$r = \sqrt{(8-5)^2 + (8-4)^2} = \sqrt{9+16} = \sqrt{25} = 5 \text{ units}$$

Radius = 5 units.

$$(P_1) = (1, 0) \quad C(4, 0) = (1, 0)$$

$$(C_1) = (4, 0) \quad (P_2) = (2, 4)$$

$$(P_1) = (1, 0) \quad (P_2) = (2, 4)$$

$$E = P_1 = (1, 0) \quad (C_2) = (2, 4)$$

$$(2, 4)$$

$$P_2 = (2, 4) \quad C_2 = (2, 4) \quad (P_1) = (1, 0)$$

$$\boxed{(2, 4)}$$